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Long Short-Term Memory-Based Fall Detection by Frequency-Modulated Continuous Wave Millimeter-Wave Radar Sensor for Seniors Living Alone

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Abstract

In this study, four types of fall detection systems for seniors living alone using x-y scatter and Doppler range images measured from frequency-modulated continuous wave (FMCW) millimeter-wave (mmWave) sensors were introduced. Despite advancements in fall detection, existing long short-term memory (LSTM)-based approaches often struggle with effectively distinguishing falls from similar activities of daily living (ADLs) due to their uniform treatment of all time steps, potentially overlooking critical motion cues. To address this limitation, an attention mechanism has been integrated. Data was collected from seven participants, resulting in a dataset of 669 samples, including 285 falls and 384 ADLs with walking, lying, inactivity, and sitting. Four LSTM-based architectures for fall detection were proposed and evaluated: Raw-LSTM, Raw-LSTM-Attention, HOG-LSTM, and HOG-LSTM-Attention. The histogram of oriented gradient (HOG) method was used for feature extraction, while LSTM networks captured temporal dependencies. The attention mechanism further enhanced model performance by focusing on relevant input features. The Raw-LSTM model processed raw mmWave radar images through LSTM layers and dense layers for classification. The Raw-LSTM-Attention model extended Raw-LSTM with an added self-attention mechanism within the traditional attention framework. The HOG-LSTM model included an additional preprocessing step upon the RAW-LSTM model where HOG features were extracted and classified using an SVM. The HOG-LSTM-Attention model built upon the HOG-LSTM model by incorporating a self-attention mechanism to enhance the model's ability to accurately classify activities. Evaluation metrics such as Sensitivity, Precision, Accuracy, and F1-Score were used to compare four architectural models. The results showed that the HOG-LSTM-Attention model achieved the highest performance, with an Accuracy of 95.3% and an F1-Score of 95.5%. Optimal self-attention configuration was found at a 2:64 ratio of number of attention heads to channels for keys and queries.

Keywords: fall detection; millimeter-wave sensor; the elderly; long short-term memory (LSTM); attention

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1. Introduction

Falls in the elderly are serious health problems with physical injuries, psychological effects, and social consequences. Falls can seriously impair the elderly's independence, increase medical costs, and lower their quality of life. According to the World Health Organization (WHO), approximately 28 to 35% of the population over the age of 65 experiences falls annually, often resulting in serious injuries or fatalities [1]. Consequently, the development of fall prevention and detection technologies is critical to ensuring the health and safety of the elderly [2].

Despite recent advancements in fall detection technologies, conventional long short-term memory (LSTM)-based models often struggle to distinguish falls from similar activities of daily living (ADLs), as they treat all time steps uniformly [3–8]. This limitation may result in the overlooking of critical motion cues, thereby increasing the likelihood of false positives and compromising system reliability in real-world applications.

This study aims to address the challenge of accurate fall detection for seniors living alone by leveraging frequency-modulated continuous wave (FMCW) millimeter-wave (mmWave) radar sensors [9,10]. These sensors offer distinct advantages over traditional methods, such as their privacy-preserving nature, robustness in varying lighting conditions, and ability to accurately capture human motion without physical contact, making them ideal for continuous and unobtrusive monitoring in private living spaces. The primary objective is to develop and evaluate robust fall detection algorithms that can effectively differentiate between various ADLs (walking, lying, inactivity, sitting) and actual falls. The scope of this research focuses on the application of x-y scatter and Doppler range images acquired from mmWave sensors.

The novelty of this study lies in its comparative analysis of four distinct LSTM-based architectures for fall detection using FMCW mmWave radar data. While LSTM with attention mechanisms has been explored with wearable and vision-based sensors [11–14], a comprehensive comparative study utilizing x-y scatter and Doppler range images from FMCW mmWave sensors has not been reported. Our research provides a detailed evaluation of how the integration of Histogram of Oriented Gradient (HOG) [15] features and an attention mechanism [16] enhances the accuracy and reliability of fall detection when applied to FMCW mmWave radar data. The identification of an optimal attention configuration further contributes to the practical implementation of such systems. The first model, the Raw-LSTM, processes unaltered mmWave radar images directly through LSTM layers. The second model, the Raw-LSTM-Attention, extended the Raw-LSTM by incorporating a self-attention mechanism within the traditional attention framework. The third model, the HOG-LSTM, enhances this by first extracting features using the HOG method [15] before passing them to the LSTM [17]. Building upon this, the fourth model, HOG-LSTM-Attention, integrates a self-attention mechanism to selectively focus on the most relevant temporal features within the attention framework [11,12,16], thereby enhancing detection accuracy.

The paper is organized as follows: Section 2 describes firstly the experimental setup and sensor configuration, detailing the office environment and the placement of the radar sensor. Next, the subject and dataset used for the experiments are introduced, including the types of activities recorded and the characteristics of the participants. The feature extraction process using the HOG method is then explained, followed by a description of the LSTM networks and the attention mechanism employed in the study. The proposed fall detection algorithms are presented, along with the hyperparameters used for the models. Finally, the evaluation metrics and the results of the experiments are discussed, comparing the performance of different architectural models for fall detection.

2. Literature Review

Various technologies for detecting falls have been developed among the elderly, each with its own advantages and limitations. Wearable devices, such as smartwatches and necklaces, often incorporate accelerometers and gyroscopes to detect sudden changes in body position and movement [3–6,18]. These sensors identify falls by measuring acceleration and rotational motion, respectively. Advanced systems use artificial intelligence (AI) and machine learning algorithms to improve accuracy and reduce false alarms [3–8]. Medical alert systems like the Medical Guardian MGMini and MobileHelp Mobile Duo offer features such as Global Positioning System (GPS) tracking, automatic fall alerts, and 24/7 monitoring, providing immediate assistance in case of a fall [19].

Unwearable systems include camera-based solutions that use artificial intelligence (AI) algorithms to detect falls in real-time. These systems provide continuous monitoring without the use of wearable devices, making them ideal for wider regions or multiple patients [8,20]. However, privacy concerns and the need for constant surveillance are significant drawbacks. Pressure sensor mats, placed on the floor, detect falls by sensing changes in pressure and are often used in combination with other systems to enhance detection accuracy [21]. In addition, hybrid approaches that integrate fall detection systems and smart home technologies are emerging [3–5]. These systems combine sensors, cameras, and wearable devices to provide comprehensive monitoring. For example, smartwatches can be paired with home security systems to alert caregivers or emergency services in case of a fall. Predictive analytics is another advanced feature in which systems analyze data from various sensors to identify fall risk patterns and prevent accidents before they happen.

Emerging technologies such as FMCW mmWave sensors are gaining attention for their high resolution and non-contact fall detection capabilities [9,10,21–26]. These sensors can penetrate clothing and other materials, providing reliable data without the need for direct contact, which addresses privacy concerns associated with camera-based systems and the inconvenience of wearable devices [9,10]. Their ability to operate in various environmental conditions (e.g., darkness, fog) and provide precise velocity, range, and angle measurements makes them particularly suitable for robust fall detection applications [9,10]. AI and machine learning algorithms are increasingly integrated with fall detection technologies, analyzing complex patterns in sensor data to improve accuracy and reliability. These different approaches highlight ongoing efforts to address the problem of fall detection in the elderly. Each technology offers unique advantages and faces certain challenges, and continuous re-research aims to improve its effectiveness and usability.

The HOG method is widely recognized for its effectiveness in object detection and recognition tasks, owing to its ability to capture the shape and appearance of objects through the distribution of intensity gradients [15]. In the domain of fall detection using mmWave radar images, HOG features are particularly advantageous, as they effectively represent the spatial patterns and structural changes associated with various human activities and fall events [27]. By analyzing gradient magnitudes and orientations within localized regions of radar images, HOG can extract robust features that are resilient to minor variations in position, scale, or illumination. This robustness enables models to identify characteristic outlines and motion dynamics of individuals, thereby enhancing the discriminative capability for distinguishing falls from ADLs [27].

Hence, various deep learning architectures have been introduced to take advantage of the rich spatial and temporal information provided by FMCW mmWave sensors. These architectures are considered robust because they can learn complex patterns and relationships within the data, which is crucial for accurately distinguishing between falls and various ADLs that may exhibit similar motion characteristics [9,10]. The ability of mmWave sensors to provide high-resolution data that is less affected by environmental factors like

lighting conditions or privacy concerns further enhances the robustness of deep learning models trained on such data [9,10]. Among them, recurrent neural networks (RNNs), especially LSTM models, have been widely used to capture temporal dependencies in human motion [24–26]. However, recent studies have revealed limitations in LSTM-only approaches, particularly in distinguishing falls from ADLs that may exhibit similar motion patterns [24–26]. Despite this over 90% accuracy success, LSTM-based models often treat all time steps equally, which can degrade the performance of identifying critical motion cues. As a solution to this drawback, an attention mechanism has been applied to improve the effectiveness of the model [28,29]. The attention mechanism allows the model to pinpoint the most important features, such as noticeable movements or posture changes, within the input sequence, which improves the accuracy of detecting falls compared to ADLs. Incorporating the attention mechanism into the LSTM architecture allows the model to weigh more critical temporal features, which improves detection accuracy and reduces false positives. This is especially important in real-world scenarios, where radar data may be noisy or where falls may resemble other rapid movements. To enhance the accuracy of fall detection, fall detection algorithms combining LSTM with attention mechanisms have been studied in wearable devices or vision-based sensors [13,14,29,30]. However, a comparative study on the performance of applying x-y scatter and Doppler range images measured by FMCW mmWave sensors using only LSTM versus combining LSTM with attention has not yet been reported.

3. Materials and Methods

3.1. Experiment and Sensor Setup

An office has been chosen as the experimental scene, as shown in the left side of Figure 1. The interior dimensions have been illustrated in the right side of Figure 1. The size of the room is 7 m × 8 m, and the radar is located at the center of the left side of the office, approximately 1.2 m above the floor. On the right side of the office, there is a full-length bookshelf and a single desk with 0.7 m × 1.4 m in front of the bookshelf.

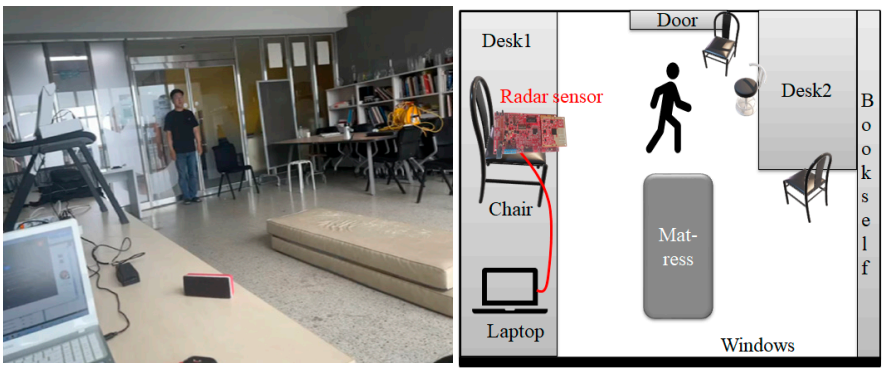

Figure 1. Real photo (**left**) and 2D layout (**right**) of experimental environment for fall detection using the mmWave radar sensor in a room.

The human tracking and fall detection experiment utilizes an FMCW mmWave radar sensor, developed by Texas Instruments (TI), along with a laptop for data processing. The radar sensor chosen for evaluation is the IWR6843ISK, which operates within a frequency range of 60 GHz to 64 GHz, offering a continuous bandwidth of 4 GHz [31,32]. It is equipped with three transmitting antennas and four receiving antennas, providing 15° azimuth and elevation coverage. The mmWave radar employs FMCW technology to precisely measure velocity, range, and angle by transmitting continuous signals with varying frequencies. The radar signals of the sensor produce various scattering points after reflecting off a human body. These points, originating from different body parts, form a unique point cloud

for each part. Typically, these points exhibit different velocities or Doppler shifts due to movements, such as leg or arm motions. This sensor captures x-y scatter data and Doppler measurements to identify detected objects, achieving a range resolution of 0.044 m and a maximum detection range of 9.02 m. The maximum measurable radial velocity is 1 m/s with a velocity resolution of 0.13 m/s. The images of x-y scatter and Doppler range plots for each activity were recorded for 10 s, generating 100 frames per activity at a sampling rate of 10 frames per second (FPS). The TI mmWave demo visualizer [33] was used to display x-y scatter plots and Doppler range plots, as shown in Figure 2, which illustrates measurements for five different activities: walking, sitting, lying down, inactivity, and falling.

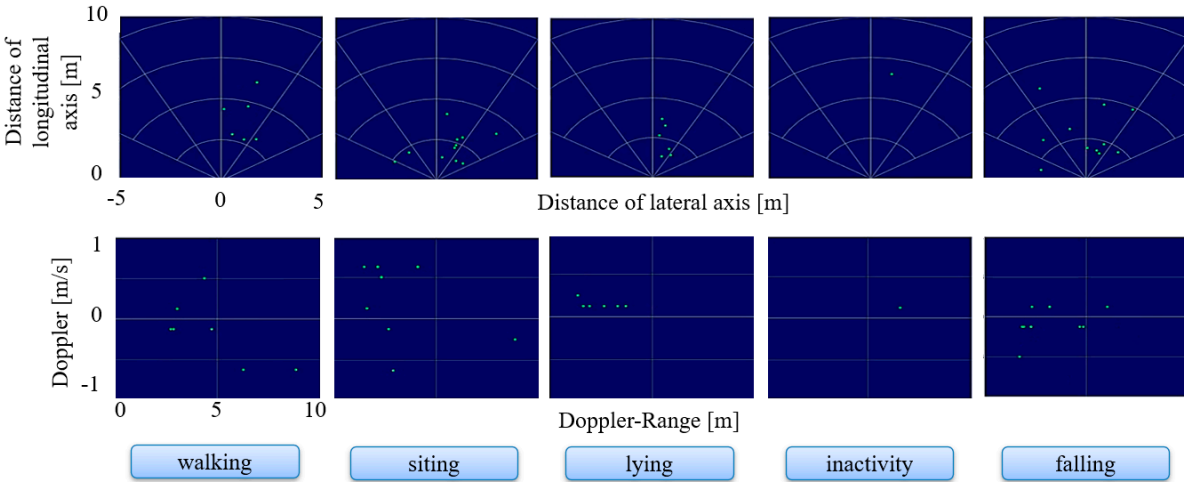

Figure 2. Images of *x-y* scatter plot (**top**) and Doppler range plot (**bottom**) of five types of activities for detected objects measured by a mmWave sensor.

3.2. Subject and Dataset

A data set consisting of falls and ADLs was acquired to detect falls from the mmWave sensor. The dataset used includes five activities, including walking, lying, sitting, and inactivity in ADL and falls, as shown in Figure 3. One dataset for each activity consists of a sequential collection of images from 100 frames, with example images shown in Figure 2, and the total dataset consists of 669 samples, with 285 representing falls and 384 corresponding to ADLs, collected from seven participants. Although the dataset includes only 669 samples, deep learning models can still be effective when applied to structured, high-quality, and temporally rich data such as radar-based motion sequences [9,10,21–26]. The radar images contain rich spatial and temporal information, which deep learning models like LSTM and attention mechanisms are well-suited to capture. Additionally, we employed techniques such as dropout regularization and careful hyperparameter tuning to mitigate overfitting. The consistent structure of radar data and the sequential nature of human motion allow LSTM-based models to generalize effectively even with limited samples. A total of seven subjects participated in the study: one female in her 20s with a body weight in the 40 kg range, two males in their 20s weighing approximately 60 kg, three males in their 20s weighing around 70 kg, and one male in his 50s also in the 80 kg weight range. Although the participants were between 20 and 50 years old, the fall scenarios were carefully crafted to realistically simulate elderly falls. This approach is supported by previous studies [34–36], which utilized simulated falls by younger adults to develop fall detection algorithms for elderly care. These studies demonstrate that fall dynamics—such as sudden changes in posture and impact patterns—are largely consistent across age groups, particularly when captured using non-contact sensors like radar. Therefore, models trained on such data remain effective in detecting falls among the elderly.

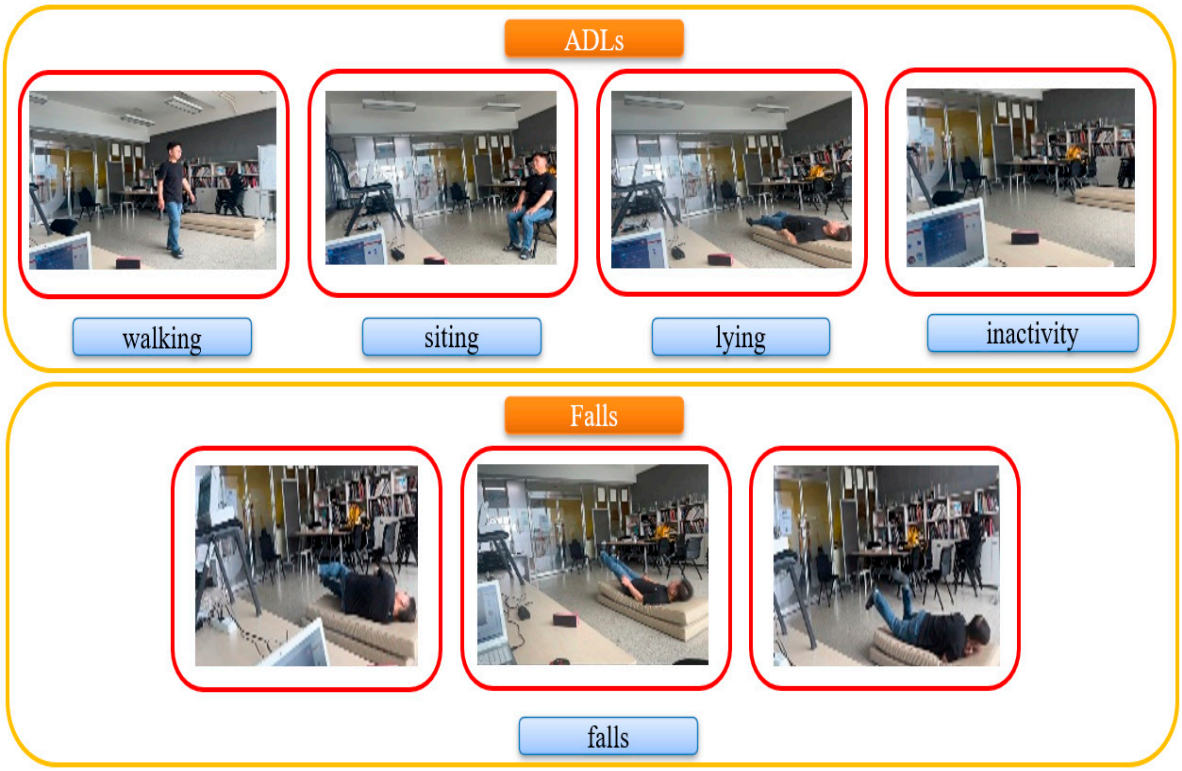

Figure 3. Photos of four types of activities of daily living (ADLs) and three types of falls for one senior.

3.3. Histogram of Oriented Gradient (HOG)

The process of extracting features using the HoG method [15] involves several key steps, as shown in Figure 4. Initially, the image is divided into small, connected regions known as cells. For each cell, the gradient of each pixel is calculated. These gradients are then used to create a histogram of gradient directions for each cell. The histograms from all cells are combined to form a comprehensive descriptor for the entire image. To analyze temporal changes, HoG features are extracted at different time intervals and compared. By recognizing patterns in the HoG features, distinct characteristics of each activity can be identified. The temporal changes in HoG features are visualized to provide insights into the dynamics of the activities. Subsequently, the extracted features can be used to train machine learning models such as a support vector machine (SVM). These models are evaluated to determine their performance and accuracy in classifying the activities based on the HoG features. This method proves to be effective in capturing the essential gradient structures that characterize different activities over time.

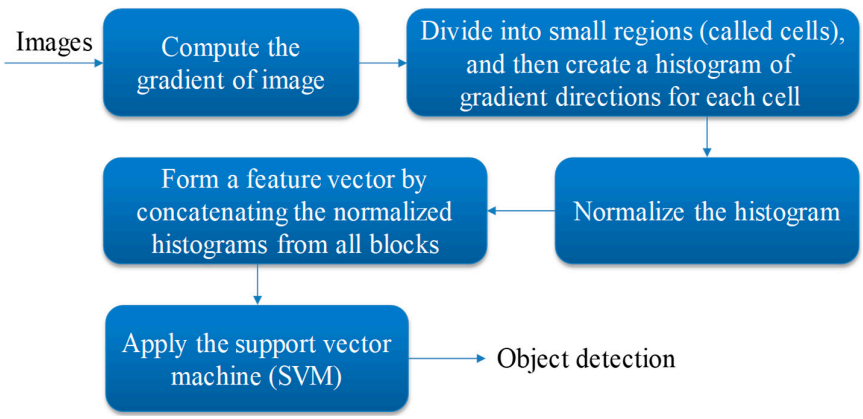

Figure 4. The procedure to extract the features of objects using HOG.

3.4. Long Short-Term Memory (LSTM)

LSTM networks are a type of RNN designed to effectively capture long-term dependencies in sequential data [17]. Unlike traditional RNNs, LSTMs are capable of learning and remembering information over extended periods, making them particularly useful for tasks involving time series data, natural language processing, and speech recognition [37–39]. The core component of an LSTM network is its memory cell, which maintains its state over time. Each memory cell is equipped with three gates: the input gate, the forget gate, and the output gate. These gates regulate the flow of information into and out of the cell, allowing the network to selectively remember or forget information. The input gate controls the extent to which new information is added to the cell state, the forget gate determines the degree to which existing information is discarded, and the output gate manages the information that is passed to the next time step. For fall detection, LSTMs are effective because they can process sequential radar data, capturing the temporal evolution of human motion patterns. This allows them to identify subtle changes over time that are indicative of a fall, distinguishing them from normal activities.

3.5. Attention Mechanism

Attention mechanisms are a powerful concept in neural networks that allow models to focus on specific parts of the input data when making predictions [11,16]. By selectively focusing on relevant information, you can improve the performance of your model. At its core, the concept works by assigning different weights to different parts of the input data. These weights determine the importance of each part, allowing the model to prioritize certain elements over others. Figure 5 shows the attention mechanism. This process involves three main components: queries ($\boldsymbol{Q}$), keys ($\boldsymbol{K}$), and values ($\boldsymbol{V}$). Queries represent the current state or parts of the input that need attention, keys represent potential parts of the input that can be attended to, and values represent the actual information that is retrieved based on the attention weights. These weights are applied to the values to create a weighted sum that represents the focused information.

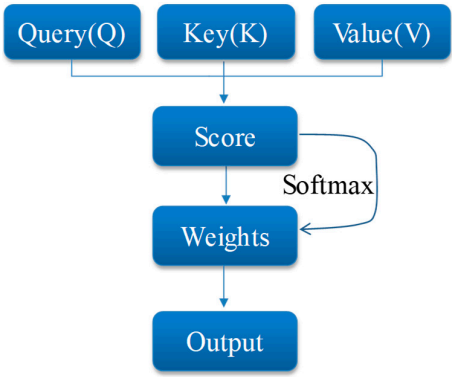

Figure 5. Attention mechanism.

Self-attention, a specific form of attention, allows the model to weigh the importance of different words or features within the same input sequence when processing each element. Unlike traditional attention, where queries, keys, and values might come from different sources (e.g., encoder and decoder in sequence-to-sequence models), in self-attention, all three components originate from the same input sequence [11,12,16]. This enables the model to capture complex internal dependencies and relationships within the input itself.

This approach allows the model to dynamically focus based on context, resulting in more accurate and context-aware predictions. In fall detection, the attention mechanism is particularly beneficial as it enables the model to pinpoint critical motion cues, such as rapid changes in posture or velocity, which are crucial for differentiating between falls and

other activities that might have similar movement profiles. This selective focus enhances detection accuracy and reduces false positives, especially in noisy radar data or scenarios where falls resemble other rapid movements.

3.6. Proposed Fall Detection Algorithms

Figure 6 shows a flow chart of the proposed fall detection algorithm. Firstly, images of x-y scatter and Doppler range plots of one object are acquired from the mmWave radar sensor. These images are then preprocessed using the HOG method, which extracts features by analyzing the gradient orientations within localized portions of the image. This preprocessing step is crucial for enhancing the quality of the data and making it suitable for further analysis. Following the preprocessing, the extracted features are fed into an LSTM network combined with a self-attention mechanism. This combination enables the model to dynamically adjust its focus based on the context, leading to more accurate and context-aware predictions. Once the features have been processed by the LSTM-Attention mechanism, they are used for classification. The classification step involves determining whether the activity depicted in the images is one of the four types of ADLs or a fall. This final step is critical for applications such as fall detection systems, where accurately identifying falls can help in providing timely assistance and improving safety.

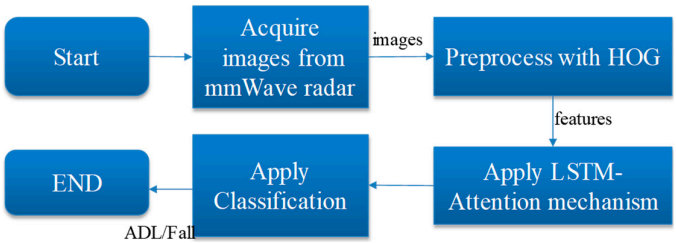

Figure 6. Flow chart of proposed fall detection algorithm.

To find the best deep learning algorithm for detecting object falls using mmWave radar images, four potential algorithms must be compared and evaluated. Figure 7 shows four types of potential architectural models designed for fall detection. The first model, Raw-LSTM, begins by taking mmWave radar images as input. The two-dimensional (2D) image frames are first converted to grayscale to reduce complexity and focus on intensity features. Each grayscale image is then reshaped into a one-dimensional (1D) array. These 1D arrays are normalized by scaling pixel values to the range [0, 1]. The resulting normalized sequential data is fed into a series of LSTM layers, each equipped with dropout regularization to prevent overfitting and improve generalization. The output from the LSTM layers is then passed through dense layers with ReLU activation functions, followed by a final dense layer with a softmax activation function for classification. This model classifies activities into categories such as sitting, walking, lying, inactivity (ADLs), and three types of falls (backward, forward, sideways). The second model, Raw-LSTM-Attention, builds upon the Raw-LSTM model by incorporating a self-attention mechanism within the traditional attention framework. A self-attention mechanism is applied to the output of the second LSTM layer, allowing the model to focus on the most relevant parts of the input data of the LSTM. This focused information is then passed through dense layers with ReLU and softmax activation functions for classification. The self-attention mechanism enhances the model's ability to accurately classify activities by dynamically adjusting its focus based on the context. The third model, HOG-LSTM, also starts with mmWave radar images but includes an additional preprocessing step where HOG features are extracted. This involves generating HOG directions within localized portions of the image, normalizing these histograms, and building a window descriptor. The extracted HOG

features are then classified using an SVM to generate feature vectors. These vectors are fed into LSTM layers with dropout, followed by dense layers with ReLU and softmax activation functions for final classification into the same activity categories as those used in the Raw-LSTM and Raw-LSTM-Attention models. The fourth model, HOG-LSTM-Attention, enhances the HOG-LSTM architecture by incorporating a self-attention mechanism to improve its ability to focus on relevant temporal features. Each of these models aims to improve the accuracy and reliability of fall detection by leveraging different preprocessing techniques and neural network architectures.

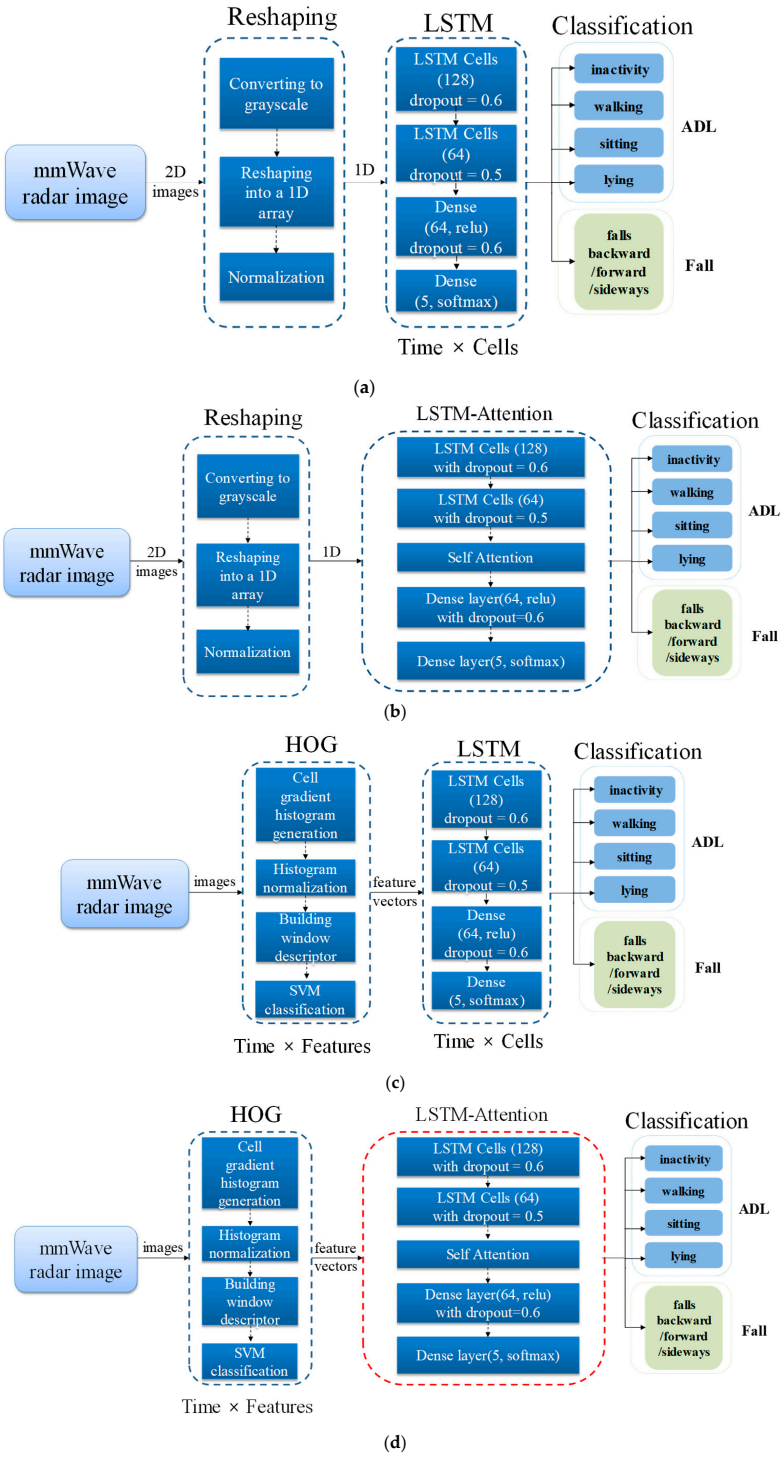

Figure 7. The four types of potential architectural models for fall detection. (**a**) Raw-LSTM, (**b**) Raw-LSTM-Attention, (**c**) HOG-LSTM, and (**d**) HOG-LSTM-Attention.

3.7. Hyperparameters of the Proposed Architecture Model

The hyperparameters of the four types of potential architectural models were kept the same using MATLAB R2025a [40] on a PC with 64 GB of RAM, an Intel(R) Core (TM) i7-14700F 2.10 GHz CPU, and an NVIDIA GeForce RTX 4090 GPU with Windows 11 Pro installed, and are shown in Table 1.

Table 1. Hyperparameters.

Names	Values/Methods
Learning rate	0.001
Epochs	1000
Batch Size	32
Optimizer	Adam
Loss Function	Crossentropy

3.8. Evaluation Metrics

To evaluate and compare the performance of the proposed fall detection algorithm with four types of potential architectural models [41], *Sensitivity* (also known as *Recall*), *Precision*, *Accuracy*, and *F1-Score* are investigated. These metrics were selected because they provide a comprehensive assessment of a model's performance, particularly in binary classification tasks like fall detection, where both false positives and false negatives have significant implications. In this study, a True Positive (TP) occurs when the model accurately identifies a fall, which is crucial for timely intervention. A True Negative (TN) is when the model accurately identifies an ADL, preventing unnecessary alarms. A False Positive (FP) is when the model incorrectly identifies an ADL data sample as a fall, leading to nuisance alerts. A False Negative (FN) is when the model incorrectly identifies a fall data sample as an ADL, which is the most critical error, as it means a real fall goes undetected. *Sensitivity*, also referred to as recall, measures the model's ability to correctly identify TPs, indicating the ratio of actual positives correctly identified by the model. In fall detection, high *Sensitivity* is paramount to ensure that actual falls are not missed. *Precision* measures the correctness of positive predictions, showing the ratio of TP predictions to all positive predictions made by the model. High *Precision* helps in minimizing false alarms, which can reduce user annoyance and caregiver burden. *Accuracy* measures the overall correctness of the model's predictions, representing the proportion of TP and TN predictions among all predictions made by the model. It provides a general overview of the model's performance. The *F1-Score* is the harmonic average of *Precision* and *Sensitivity*, offering a balanced evaluation of the model's performance by considering both *Precision* and *Sensitivity*. This metric is especially important in datasets with imbalanced classes, as it provides a more robust measure than *Accuracy* alone, ensuring that the model performs well on both falls and ADLs.

4. Result

The data is divided into 80% and 20% for training and testing, respectively. First, we predicted accuracy using the fall detection dataset with the Raw-LSTM model without any pre-processing, such as HOG, as shown in Figure 7a, the Raw-LSTM-Attention model as shown in Figure 7b, the HOG-LSTM model as shown in Figure 7c, and the HOG-LSTM-Attention model as shown in Figure 7d. Finally, we compared the performance of four types of proposed fall detection algorithms with RAW-LSTM, Raw-LSTM-Attention, HOG-LSTM, and HOG-LSTM-Attention.

Figure 8a–d show the confusion matrices of the validation data, obtained from the RAW-LSTM, Raw-LSTM-Attention, HOG-LSTM, and HOG-LSTM-Attention architectures for the fall detection system, respectively. It presents four confusion matrices for differ-

ent fall detection systems, comparing the actual labels with the predicted labels for five activities: falling, lying, no-activity, sitting, and walking. In Figure 8a, the RAW-LSTM model demonstrates significant limitations in distinguishing between different activities. While it correctly predicts all instances of falling (100% TPs for falling), it severely misclassifies all instances of ADL as falling. This results in a high number of FPs for falls and zero TNs for ADLs, indicating that the model is unable to differentiate between these activities effectively. The model essentially acts as a fall detector that flags any activity as a fall, rendering it impractical for real-world applications due to an unacceptable false alarm rate. In Figure 8b, the Raw-LSTM-Attention architecture demonstrates performance metrics comparable to those of the Raw-LSTM model, suggesting that the incorporation of the attention mechanism does not yield significant improvement. This observation is consistent with the results presented in Figure 8a. In Figure 8c, the HOG-LSTM model demonstrates an improvement in the model's performance. The model correctly predicts 94.5% of falling instances, with a small percentage (5.5%) misclassified as lying. For lying, the model correctly predicts 50% of instances, but it still misclassifies 44.4% as falling and 5.6% as sitting. The model performs exceptionally well in identifying no-activity, with a 100% correct prediction rate. Sitting is correctly predicted 95% of the time, with a small misclassification rate of 5% as lying. Walking is also accurately predicted 97.3% of the time, with 2.7% misclassified as sitting. Overall, the HOG-LSTM model significantly reduces FPs for falls compared to Raw-LSTM and shows strong performance for distinct ADLs like no-activity, sitting, and walking. Figure 8d shows the best overall performance among the four models. The HOG-LSTM-Attention model correctly predicts 98.2% of falling instances, with only 1.8% misclassified as lying. For lying, the model correctly predicts 61.1% of instances, with 38.9% misclassified as falling. The relatively low classification accuracy for the lying activity, as shown in Figure 8c,d, can be attributed to its visual and motion similarity with the falling activity in radar-based detection systems. Both activities often result in similar horizontal postures and minimal movement, which can confuse the model, especially when relying on Doppler range and scatter plots [9,10]. The model maintains a high accuracy for no-activity, correctly predicting 97.3% of instances, with a small misclassification rate of 2.7% for sitting. Sitting is accurately predicted 98.2% of the time, with 1.8% misclassified as walking. Walking is also correctly predicted 97.3% of the time, with 2.7% misclassified as sitting. In summary, Figure 8a,b illustrate the model's difficulty in distinguishing falls from four types of ADLs, resulting in a high rate of misclassification. Figure 8c shows improved performance, particularly in identifying no-activity, sitting, and walking, but it still has issues with misclassifying lying instances. Figure 8d exhibits the best overall performance, with high accuracy in predicting falling, no-activity, sitting, and walking, though it still faces some challenges with lying instances. Overall, the HOG-LSTM-Attention model appears to be the most effective model for fall detection, providing the highest accuracy across most activities.

Actual Label \ Predicted Label	falling	lying	inactivity	sitting	walking
falling	1.00	0.00	0.00	0.00	0.00
lying	1.00	0.00	0.00	0.00	0.00
no-activity	1.00	0.00	0.00	0.00	0.00
sitting	1.00	0.00	0.00	0.00	0.00
walking	1.00	0.00	0.00	0.00	0.00

(**a**)

Figure 8. *Cont.*

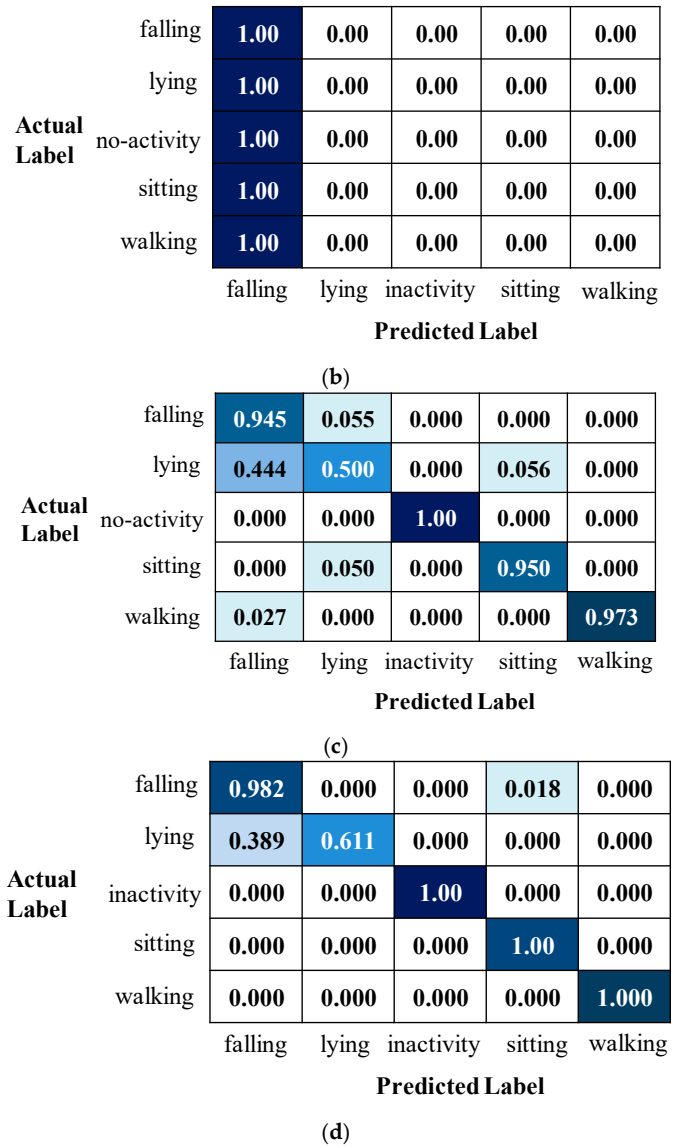

Figure 8. Confusion matrix applying the four types of proposed fall detection systems. (**a**) Raw-LSTM, (**b**) Raw-LATM-Attention, (**c**) HOG-LSTM, (**d**) HOG-LSTM-Attention.

Table 2 shows the *Sensitivity*, *Precision*, *Accuracy*, and *F1-Score* obtained from the confusion matrix of the four types of fall detection systems shown in Figure 8. The Raw-LSTM architecture demonstrates perfect *Sensitivity*, correctly identifying all instances of falls (100%). However, its *Precision* is significantly lower at 50%, indicating that the model struggles to correctly identify ADLs, leading to a high rate of FPs. The overall *Accuracy* of this model is also 50%, reflecting its balanced but mediocre performance, which is heavily influenced by the high number of false alarms. The *F1-Score*, which is the harmonic mean of *Precision* and *Sensitivity*, stands at 66.7%, suggesting that while the model is excellent at detecting falls, it is less reliable in distinguishing between falls and ADLs. The Raw-LSTM-Attention architecture yields performance metrics equivalent to those of the Raw-LSTM model, indicating no significant improvement from the addition of the attention mechanism. The HOG-LSTM architecture shows a marked improvement in performance. Its *Sensitivity* is slightly lower than the Raw-LSTM and Raw-LSTM-Attention models at 94.5%, but it still performs well in detecting falls. The *Precision* is significantly higher at 88.9%, indicating that the model is much better at correctly identifying ADLs and reducing FPs. The overall *Accuracy* of this model is 91.4%, demonstrating a balanced and reliable performance. The *F1-Score* is 91.6%, reflecting the model's strong ability to detect falls while maintaining a

high level of *Precision*. The HOG-LSTM-Attention architecture exhibits the best overall performance among the four models. It achieves a high *Sensitivity* of 98.2%, indicating that it is very effective at detecting falls. The *Precision* is also high at 92.9%, showing that the model is proficient at correctly identifying ADLs and minimizing FPs. The overall *Accuracy* of this model is 95.3%, reflecting its superior performance in both detecting falls and distinguishing between falls and ADLs. The *F1-Score* is 95.5%, indicating a well-balanced and highly reliable model, benefiting from the attention mechanism's ability to focus on critical temporal cues. In summary, the Raw-LSTM and Raw-LSTM-Attention models excel in *Sensitivity* but fall short in *Precision* and *Accuracy*. The HOG-LSTM model offers a more balanced performance with high *Sensitivity* and *Precision*. The HOG-LSTM-Attention model provides the best results across all metrics, making it the most effective architecture for fall detection by combining robust feature extraction with dynamic temporal focus.

Table 2. Validation results of fall detection of four types of proposed fall detection systems.

Architecture	Evaluation Metrics [%]			
	Sensitivity	***Precision***	***Accuracy***	***F1-Score***
Raw+LSTM	100	50	50	66.7
Raw+LSTM+Attention	100	50	50	66.7
HOG+LSTM	94.5	88.9	91.4	91.6
HOG+LSTM+Attention	98.2	92.9	95.3	95.5

Table 3 presents various evaluation metrics for different ratios of the number of Attention heads (NumHeads) to the number of channels of keys and queries (NumKeyChannels) of self-attention mechanism in a HOG-LSTM-Attention model for the fall detection system. These metrics include *Accuracy*, *Sensitivity*, *Precision*, and *F1-Score*, each expressed as a percentage. The *Accuracy* metric measures the overall correctness of the model's predictions. The highest *Accuracy* is observed at the 2:64 ratio, with a value of 95.3%, indicating that this configuration provides the most accurate predictions. As the ratio increases, the *Accuracy* generally decreases, with the lowest *Accuracy* recorded at the 32:1024 ratio, which is 85.0%, suggesting a diminishing return or even a detrimental effect with excessive attention complexity. The highest *Sensitivity* is seen at the 2:64 and 4:128 ratios, both achieving 98.2%. This suggests that these configurations are particularly effective at identifying TPs, crucial for not missing actual falls. *Sensitivity* drops significantly at the 32:1024 ratio, reaching a low of 76.4%, indicating that overly complex attention mechanisms might diffuse focus rather than enhance it. The highest *Precision* is found at the 2:64 ratio, with a value of 92.9%, demonstrating that this configuration has the most accurate positive predictions. *Precision* varies across different ratios, with the lowest *Precision* at the 1:32 ratio, which is 88.7%, indicating that insufficient attention heads or channels might lead to more FPs. The highest *F1-Score* is observed at the 2:64 ratio, with a value of 95.5%, indicating that this configuration offers the best balance between *Precision* and *Sensitivity*. The *F1-Score* decreases at higher ratios, with the lowest value at the 32:1024 ratio, which is 83.5%, reinforcing the notion that the 2:64 ratio provides the optimal trade-off between accurately detecting falls and minimizing false alarms. Overall, the 2:64 ratio appears to be the most effective configuration across all metrics, providing the highest *Accuracy*, *Sensitivity, Precision*, and *F1-Score*. As the ratio of attention heads to key channels increases beyond this optimum, the performance of the model generally declines, suggesting that an excessively complex attention mechanism can lead to overfitting or an inability to generalize effectively.

Table 3. Evaluation metrics for different numbers of attention heads (NumHeads) and numbers of channels for keys and queries (NumKeyChannels) ratios for fall detection system with HOG-LSTM-Attention model.

Evaluation Metrics	NumHeads:NumKeyChannels					
	1:32	2:64	4:128	8:256	16:512	32:1024
Sensitivity [%]	92.7	98.2	98.2	94.5	94.5	76.4
Precision [%]	88.7	92.9	90.1	90.7	88.9	92.2
Accuracy [%]	90.4	95.3	93.7	92.4	91.4	85.0
F1-Score [%]	90.7	95.5	94.0	92.6	91.6	83.5

5. Conclusions

This study focused on developing and evaluating a fall detection system using mmWave radar technology. The experimental setup involved placing the radar sensor in an office environment to monitor human activities and detect falls. The radar sensor used captured *x*-*y* scatter data and Doppler measurements to identify and classify different ADLs and falls. The dataset for the experiments included five activities: walking, lying, sitting, inactivity, and falls. The data was collected from seven participants, resulting in a total of 669 samples, including 285 falls and 384 ADLs.

Key findings of this research highlight the superior performance of the HOG-LSTM-Attention model. The architecture, which combines HOG for feature extraction with a self-attention mechanism integrated into LSTM networks, consistently outperformed the Raw-LSTM, Raw-LSTM-Attention, and HOG-LSTM models across all evaluation metrics. Specifically, the HOG-LSTM-Attention model achieved an *Accuracy* of 95.3%, *Sensitivity* of 98.2%, *Precision* of 92.9%, and an *F1-Score* of 95.5%. The optimal self-attention configuration was identified as a 2:64 ratio of attention heads to channels for keys and queries, demonstrating the importance of fine-tuning these parameters for peak performance. These results affirm that combining robust feature engineering (HOG) with advanced temporal modeling (LSTM) and selective focus (attention) significantly enhances the accuracy and reliability of fall detection using mmWave radar.

Despite these promising results, several limitations were encountered during this study. Firstly, the dataset, while diverse in activities, was collected from a relatively small number of participants (seven individuals aged 20–50). This demographic may not fully represent the diverse movement patterns and physical conditions of the elderly population (over 65 years old), who are the primary beneficiaries of such systems. Therefore, the generalizability of the model's performance to a broader, older demographic might be limited. Secondly, the experiments were conducted in a controlled office environment. Real-world living spaces can vary significantly in layout, presence of obstacles, and background clutter, which could affect radar signal propagation and detection accuracy. Thirdly, while the model showed high overall accuracy, distinguishing between 'lying' and 'falling' activities remained a challenge, as indicated by persistent misclassifications in the confusion matrices. This suggests that certain ADLs can still mimic fall patterns, leading to potential false alarms or missed detections.

Future research should focus on expanding dataset diversity by including a wider range of participants, particularly older adults with varying body types and mobility profiles, to enhance model generalization across real-world populations. Additionally, testing in more realistic and heterogeneous home environments—with varied spatial layouts and interference sources—is essential to evaluate performance under operational conditions.

The findings from this study strongly suggest that mmWave radar technology, when combined with advanced machine learning techniques like HOG feature extraction and attention-enhanced LSTM, can be effectively used for highly accurate and reliable fall detection. By addressing the identified limitations, this technology has the potential to significantly contribute to improved safety and independent living for vulnerable populations.

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